

Experiment no-12

12. Implement an Earley parser for context-free grammars using a simple python program.

PROGRAM:

```
import nltk
from nltk import CFG
from nltk.parse import EarleyChartParser

grammar = CFG.fromstring("""
S -> NP VP
NP -> Det N
VP -> V NP
Det -> 'the' | 'a'
N -> 'cat' | 'dog'
V -> 'chased' | 'saw'
""")

parser = EarleyChartParser(grammar)

sentence = input("Enter a sentence: ").lower().split()

parsed = False

for tree in parser.parse(sentence):
    print(tree)
    parsed = True

if not parsed:
    print("Sentence cannot be parsed.")
```

OUTPUT:

```
*** Enter a sentence: the cat chased a dog
(S (NP (Det the) (N cat)) (VP (V chased) (NP (Det a) (N dog))))
```